Multilayer Perceptrons (MLPs)

Multi-Layer Perceptrons (MLPs) are *feedforward networks* in which information flows in one direction: starting from the input layer, then through one or more hidden layers, and lastly the output layer

Consist of fully connected laters where each neuron connects to every neuron in the next layer.

Non-linear activation function transformations are applied as the data flows through multiple hidden laters and the network learns complex patterns.

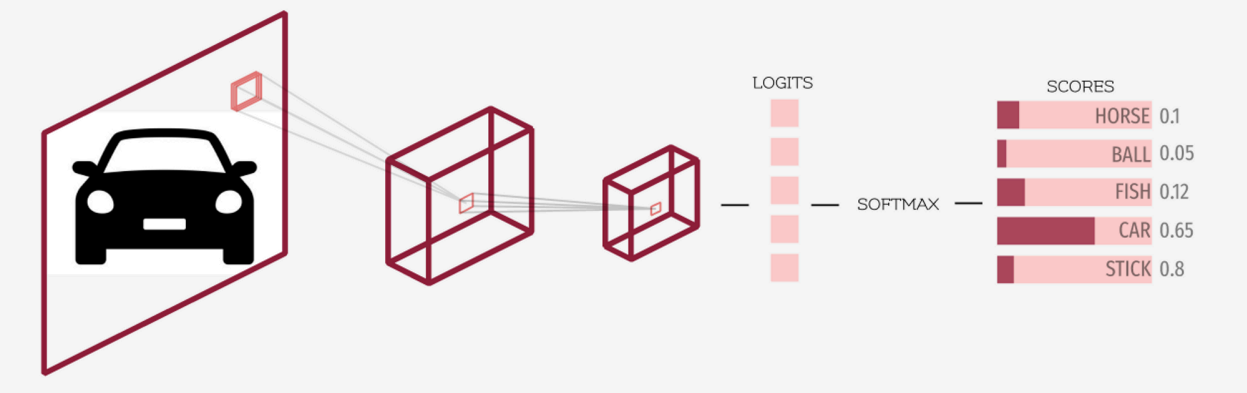
The most widely used activation function is the **Rectified Linear Unit (ReLU)** activation function, which zeros out negative values while keeping positive values.

Universal Approximation: an MLP with at least one hidden layer and a non-linear activation function can approximate any continuous function mapping the inputs to the target output.

Training an MLP to learn the relationship between inputs and outputs involves three components:

1. Loss Function: measures how far the network's predictions are from the actual targets
2. Optimizer: A method for adjusting the model's parameter weights during training to reduce the loss function.
3. Training Loop: iterations (epochs) over batches of the training dataset, where in each iteration we:
    a. Pass the input batch through the network's forward pass;
    b. Obtain the raw outputs (logits) and compute the loss;

c. Perform a backward pass (backpropagation);

d. Use optimizer to update the weights to reduce the loss;

e. Track training metrics like the average loss.